Breaking SAM 2: An Evaluative Benchmark on Extreme Object Evolution

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Abstract

Standard video segmentation benchmarks mostly contain objects with a stable shape and appearance. We present a small, inference-only stress test that measures *when*, *how*, and *under which transformation* SAM 2 fails, on 52 clips from M3-VOS, VOST, DAVIS 2017, and VOT: 14 with topology/state change, 14 with extreme scale or viewpoint change, 10 with illumination change, and 14 controls. We compare SAM 2 prompted with a verified box to Grounded-SAM 2, where Grounding DINO gives the first box from a text prompt, and report detector misses separately from propagation failures. With a verified box, SAM 2 fails on 12 of 14 topology clips (median 2.0 s after the event) and 5 of 10 illumination clips, but on only 1 of 14 scale clips and no control clip. Grounding DINO fails to initialize on 11 of 52 clips; on the other 41, Grounded-SAM 2 is only 2.0 $J\&F$ points below SAM 2. Code: https://github.com/YarinBS/breaking-sam2.

1 Introduction

SAM 2 [1] extends the promptable Segment Anything Model [2] to video: the object is prompted once, and a streaming memory of the prompted frame and of recent predictions is used to segment every new frame. In Grounded-SAM 2 [4], a box from the open-set detector Grounding DINO [3] for a free-text phrase replaces the manual prompt. Uses such as dataset labeling and robot manipulation often contain objects that do not behave like the rigid objects of standard benchmarks: vegetables are cut, water is poured, a car grows from a few pixels to half of the frame, the lighting changes within one frame. SAM 2's memory assumes that the target keeps similar visual cues over time, and amorphous targets such as "water" are hard to describe with one box. We ask *when, how, and under which transformation* SAM 2 fails, keep the two error sources of Grounded-SAM 2 (a wrong first box, later drift) apart, and study: (i) which disruption causes the earliest and most severe degradation; (ii) whether failures are abrupt, gradual, or recoverable; (iii) how sensitive propagation is to the initial prompt; and (iv) whether text-guided initialization helps or only adds a first-frame detection failure.

Related work. DAVIS [5, 6] and YouTube-VOS [7] defined the standard region (J) and boundary (F) metrics but mostly contain objects with a stable identity; VOST [10] and M3-VOS [8] target objects that change state or phase. We test a promptable *foundation* model and its text-initialized variant, add scale/viewpoint and illumination changes, and report *temporal* failure behavior, not only mean accuracy.

2 Method

Benchmark. Each category breaks a different assumption of SAM 2's memory: that the object keeps its form, its size, or its appearance (Fig. 1). **Topology/state** (14 clips; M3-VOS 8, VOST validation 6): the object is cut, broken, or torn, it flows, melts, or it is smoke. **Scale/viewpoint** (14): the object's size changes by at least 5× (up to 170×), or the view rotates strongly. **Illumination** (10): mostly abrupt light changes (muzzle flashes, stage lights, darkness), three gradual. **Control** (14): ordinary motion, nearly constant size and lighting, one object of the target's class. Scale, illumination, and control clips come from DAVIS 2017 (11, 3, 11) and VOT-ST2022 [9] (3, 7, 3). All clips (1.5–35 s) have expert per-frame ground truth and were fixed before any model was run on them. Each clip has a fixed text prompt and an event frame t_{ev}, checked by eye: the first lasting split or area change (topology), the first 5× area change (scale), or the start of the light change (illumination).

Pipelines. Both conditions use SAM 2.1 Hiera-B+ and propagate from frame 0 to the end. (A) **SAM 2**: a manually verified box (tight around the frame-0 ground truth) isolates propagation errors. (B) **Grounded-SAM 2**: the top box of Grounding DINO (tiny; box/text thresholds 0.35/0.25) on frame 0 prompts SAM 2; without a detection the masks are empty. This, and any run with $J_0 < 0.5$, is an *initialization failure*, not a propagation failure. (C) **Prompt sensitivity**: (A) with the box shifted right and down by 10% of its size, and enlarged by 20% around its center.

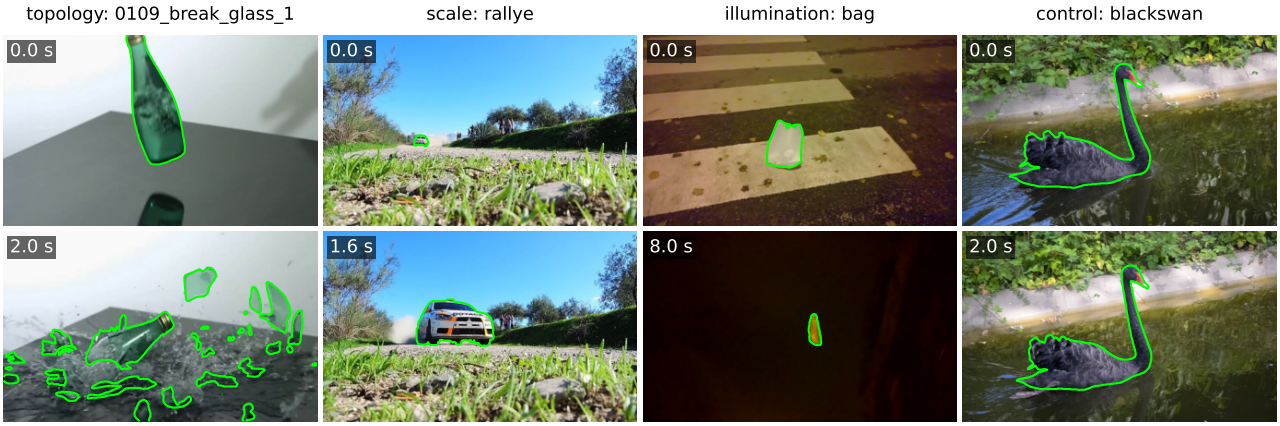


Figure 1: One clip per category, ground truth in green. Top: frame 0 (prompted). Bottom: a later frame after the event.

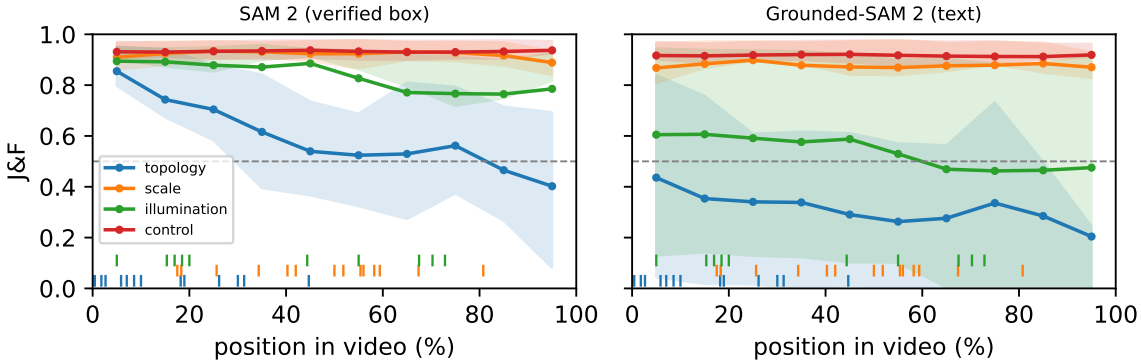


Figure 2: Mean $J\&F$ (band: IQR over clips) against the position in the clip (0% = prompted frame); ticks mark each clip's event. Left: SAM 2. Right: Grounded-SAM 2, including initialization failures ($J\&F = 0$).

Metrics. For each annotated frame t with prediction P_t and ground truth G_t: $J_t = |P_t \cap G_t|/|P_t \cup G_t|$, the boundary F-measure F_t (tolerance 0.8% of the image diagonal), and $J\&F = (J + F)/2$; to explain *how* a clip fails also the signed area error, the number of connected components of P_t, and the temporal IoU between P_{t-1} and P_t. The pre-registered first failure is

$$t_{\text{fail}} = \min\{\, t_i : J_{t_i} < 0.5 \ \wedge \ J_{t_{i+1}} < 0.5 \,\}, \qquad \Delta = (t_{\text{fail}} - t_{\text{ev}})/\text{fps}. \tag{1}$$

Clips without such a run are *censored*; controls use $t_{\text{ev}} = 0$. Recovery is a later run of two frames with $J \geq 0.5$. A failure is *abrupt* when the temporal IoU at t_{fail} is below 0.5 and *gradual* otherwise.

3 Results

All runs use one NVIDIA A100 GPU, PyTorch 2.14, SAM 2 (commit `2b90b9f`), and Transformers 4.50; `python reproduce_results.py` downloads the data, runs all conditions, and prints every number below. On another GPU (an NVIDIA H200) the means changed by at most 0.2 $J\&F$ points; all failure counts and median Δ values stayed the same.

Degradation across the video. With the verified box, all categories start at a similar level (mean $J\&F$ 0.85–0.93 in the first 10% of the clip; Fig. 2), so the later differences come from propagation. Topology clips degrade steadily after their early events (median at 9% of the clip) to 0.40 at the end; illumination clips drop after their events (median at 32%) from 0.89 to 0.79; scale and control clips stay at about 0.9. With Grounded-SAM 2, topology and illumination start much lower (0.44 and 0.61), and the gap to SAM 2 does not grow later.

Which disruption breaks SAM 2 first? (RQ i) Topology/state change causes the earliest and most frequent failures: 12 of 14 clips fail, with a median Δ of 2.0 s (Table 1, Fig. 3). Illumination change is second: 5 of 10 clips fail (median 1.1 s), e.g. `bag` after it enters near darkness and `shooting` after the muzzle flashes, while gradual light changes cause no failure. No control clip fails. Surprisingly,

		SAM 2 (box)			Grounded-SAM 2 (text)			
Category	Clips	$J\&F$	Fail	$\tilde{\Delta}$	$J\&F$	Fail	$\tilde{\Delta}$	Init
Topology	14	59.5	86%	2.0	31.2	86%	1.9	50%
Scale	14	92.1	7%	9.5	87.8	7%	−4.0	0%
Illumination	10	83.3	50%	1.1	53.8	33%	0.8	40%
Control	14	93.3	0%	–	91.7	0%	–	0%
All	52	82.0	35%	2.0	67.1	22%	1.1	21%

Table 1: Fail: share of clips with a first failure; $\tilde{\Delta}$: median time from event to failure (s); Init: Grounded-SAM 2 initialization failures. Grounded-SAM 2 Fail and $\tilde{\Delta}$ use initialized clips only.

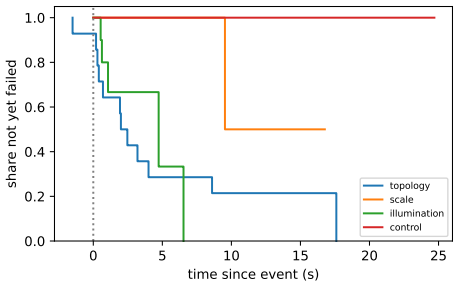


Figure 3: Kaplan–Meier survival until the first failure of SAM 2 (verified box), from the event.

only 1 of 14 scale clips fails (`helicopter`, 9.5 s after the event, at 0.05% of the image); objects that grow up to 170× are tracked without failure.

Abrupt, gradual, or recoverable? (RQ ii) Of the 18 SAM 2 failures, 9 are abrupt and 9 gradual (topology: 5 of 12 abrupt). 13 of 18 failed clips later have two consecutive frames with $J \geq 0.5$ again, but this is not good tracking: the mean worst-frame J of topology clips is 0.20 (0.86 for controls), and their masks have twice as many connected components (2.7 vs. 1.3).

Prompt sensitivity (RQ iii). Shifting the verified box by 10% of its size lowers the mean $J\&F$ by 5.5 points and makes failures 1.4 s earlier (on clips that fail in both runs); enlarging it by 20% costs 6.6 points and 1.1 s. Each change flips the failure status of 3 of 52 clips.

Initialization vs. propagation (RQ iv). Grounding DINO fails to initialize on 11 of 52 clips: 7 of 14 topology and 4 of 10 illumination clips, but no scale or control clip. The prompts name amorphous or small targets, and scenes often contain several objects of the same class: on `0018_cut_pepper_5` the top box (score 0.70) covers the whole bunch of peppers, while the second box (0.41) would have fit the target. On the 41 correctly initialized clips, Grounded-SAM 2 is 2.0 $J\&F$ points below SAM 2, almost all from three clips with a looser detector box (`helicopter`, `soapbox`, `hike`). Text initialization therefore mainly adds a first-frame failure mode.

4 Discussion

SAM 2 is robust to large changes of scale and viewpoint and to gradual lighting changes, but it breaks when the object itself changes (failures in 86% of topology clips, about 2 s after the event) and, in half of the illumination clips, under abrupt light changes. When J falls while the temporal IoU stays high, the mask is stable but wrong: the memory keeps the old appearance while the object changes; when both drop, the target is lost suddenly. For practical systems this suggests re-prompting or refreshing the memory after a large change of the object, and checking the first-frame detection before trusting a text-initialized track.

Limitations. The benchmark is small, and clip selection may add bias. VOST is sampled at 5 fps; clip lengths differ between datasets; event times are estimated from the ground truth; several VOT clips combine a light change with scale change. An overlap of SAM 2.1's undocumented training data with DAVIS cannot be excluded. All results are for one checkpoint per model.

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